3

Cost-Effective Mining Schemes

What constitutes a cost-effective mining scheme? Once two or more alternatives are available, selection of a cost-effective mining scheme implies some objective measure of comparison. Simple rules such as minimum operating cost or maximum internal rate of return often prove inadequate for comparisons of alternatives with different capital requirements or different risks. The first part of this chapter examines some of the key elements that characterize different mining schemes from this perspective.

However, there is an even more fundamental element in the quest for a cost-effective mining scheme. This element is concerned with how the alternatives chosen for evaluation come to be considered in the first place. A systematic method of evaluation also requires a mechanism to discover these possible alternatives. This systematic process of discovery, coupled with refinement of rules, is addressed in the second part of this chapter. The final part of the chapter looks at the various phases of evaluation and how the reliability, return on investment, and risk change with increasing refinement of a mine plan.

KEY ELEMENTS

For initial evaluation, the key elements of any cost-effective mining scheme can be categorized into four groupings:

- operating cost characteristics
- capital requirements and capital characteristics
- sensitivity to change and scope to adapt to change
- consistency with knowledge and philosophy of the owner

Operating Cost Characteristics

Low operating costs are always a desirable objective. If a given company has lower operating costs than its competitors, then for the same mine output selling at the same market price, the company will have larger cash flows. This gives the company greater protection against variability in market price, and it means that the company can continue in production at prices that its competitors cannot match. The company may be incurring losses, but the competitors will be losing more money.

An objective of low operating costs is particularly important for export-oriented mines, since the price of mineral commodities sold on the export market is usually more volatile than that of similar commodities supplied to domestic markets. Indeed, some domestic contracts use "cost plus" pricing, in which case the incentive might be to increase operating costs.

Low operating costs can come about because the deposit is better (shallower, higher grade) than competitive deposits. Apart from a corporate objective to seek out such deposits, this characteristic is largely outside the control of the mine planner. Low operating costs can also come about through more capital-intensive mining schemes, but this route has a price in terms of higher capital costs or perhaps less flexibility. In such cases the choice is far from clear-cut.

Capital Requirements and Capital Characteristics

Issues relating to capital fall into three groupings: capital-intensiveness, timing of capital expenditures, and the amount of capital.

Capital-Intensiveness

The term capital-intensive is used to describe investments for which there is additional expenditure of resources now with the expectation of lower costs later. It implies that, in operation, fixed costs make up a large component of total costs. When practitioners talk about implementing a mine plan that is more capital-intensive than some alternative plan, they are usually referring to the expenditure of (more) capital at the start of a mine to achieve low or lower operating costs throughout the mine life. Sometimes additional capital is expended to achieve a reduced risk or to set up the mine for easier expandability. Capital intensification is a sought-after objective in most large-

scale mining in the world, but schemes that are more capital-intensive are not necessarily desirable, nor are less capital-intensive schemes necessarily undesirable. A higher capital cost may mean a disproportionate exposure (financial risk) in unstable political environments. Capital-intensive schemes are frequently less flexible than alternatives, and this may result in higher technical risk or less ability to change with changes in the market.

Delayed Capital Expenditure

If all other things are equal, delayed capital expenditure is a desirable objective, since capital incurred later has a lower cost (in present value terms) than the same expenditure incurred today. Nevertheless, all other things may not be equal. Higher operating costs incurred by delaying capital might mean that a company is foregoing a return that may exceed the return they are getting on their money elsewhere. If properly incorporated into the mine plan, delayed capital commonly allows three other advantages:

1. The characteristics of the mine will be better understood at the time of commitment. Less capital may be required.
2. Delayed capital is subject to less uncertainty, implying a lower cost than if all of the capital were expended initially, when knowledge of the mining conditions is less well understood.
3. Even if the total capital is unchanged in present value terms, it can be partly funded from cash flow. Exposure--the maximum amount of cash outflow needed before cash starts flowing back from the investment--is reduced, and financing costs may be lower.

Capital Expenditure Appropriate to the Structure of the Company

Whenever a new proposition for investment is advanced, one of the first questions asked is, How much will it cost? The implication is that capital costs should be minimized. This implication is unfortunate. Although the owners of companies are sensitive to any proposition for capital expenditure, the raison d'être of an "investment" is to "use" capital. Efficient deployment of a firm's resources (including capital) is certainly important, but minimization of capital per se is not a key

element in the selection of a cost-effective mining method.

What is important is a total capital requirement consistent with the size of the company. A medium-sized mining company may be able to afford $200 million of investment comfortably from existing resources (retained earnings, established lines of credit), but beyond this it might have to raise additional equity or use more costly forms of finance. The extra cost of this extra capital might be very high--particularly if it puts at risk other established businesses of the company. The marginal cost of capital as a guideline for investment is discussed at length in Chapter 14.

Sensitivity to Change and Scope to Adapt to Change

The future never materializes according to any one plan or set of plans, yet shareholder value is directly related to how consistently businesses perform, year in and year out, despite changes occurring around them. This robustness in the face of change may be an inherent characteristic of a deposit, but more often it is a function of (1) the mining method and equipment chosen, (2) whether this equipment or method is adaptable or not, and (3) what institutional constraints are placed on adaptation.

Sensitivity to Influences Outside of the Company's Direct Control

In most mining applications, the key factors impacting mine economics that are outside the direct control of the company are the market price and volume, exchange rates, and mineral deposit characteristics. If waste hardness is unknown, for example, a mining scheme for which the productivity and costs are relatively insensitive to this factor is preferred. The cost of truck/shovel schemes, for example, is not as sensitive to waste hardness as is the cost of bucket-wheel excavator schemes. It may be preferable to adopt an alternative, less sensitive scheme even if this scheme has a higher cost. The cost of removing the uncertainty must be balanced against the higher cost of the less sensitive scheme.

At the start-up of a mine, many deposit characteristics may be quite unknown--favoring initial mining schemes that are less sensitive to these characteristics (even at higher costs, initially). Once the mine has been operating for some time,

these characteristics may become better understood, and the bounds of predictability may increase. Mining schemes that are sensitive to these characteristics can be implemented if they offer cost advantages within these bounds. The mine plan should also recognize the need for later change.

Scope for Production Variation

The biggest unknown in many mining projects is the sale price and offtake of the company's products. For this reason, almost all new mines commence production at annual rates below their theoretical optimum rate. If the mine were to commence production at a higher rate initially, some or all of the production might have to be sold into less-developed markets with a low marginal return. If production is increased as markets expand or as older mines close, consistency in market pricing can be maintained. A more cost-effective mine plan is one that includes within its scope these production increases (and perhaps production decreases toward the end of mine life) in line with market requirements.

Economic Limits Unconstrained

Within the economic life of a mine, enormous changes always occur in the efficiency of mining and transport and in the price of the mine's product. The establishment costs and disestablishment costs are very substantial, and frequently the most profitable period occurs after the mine is more than 10 years old. At this time the mine might even be exploiting reserves that were considered to be quite uneconomic at the start. Although doing so may be difficult to quantify on a discounted cash flow basis, mines should be designed wherever possible to allow this ease of mine extension. Major crushing facilities should be placed above mining areas that have the lowest probability of ever being mined. Alternatively, the present value of cost savings for closer placement of surface facilities should be clearly documented, so that when change occurs in the future, operators then will be alert to the economic logic that was originally employed; this reasoning can be overruled if it proves to be no longer relevant.

Consistency With Knowledge and Philosophy of the Owner

Each year large numbers of mining properties change hands, and under new ownership there are frequently quite dramatic

changes in mine performance. Equipment productivity calculations sourced from statistics or manufacturers' handbooks frequently overlook the fact that a machine produces some "rated" output only under the guidance of a skilled operator. The skills of this operator, the instructions for the task, and the ease with which other mine operations mesh with the task at hand all play a role in achieving expected outputs. What differentiates mining from almost all other industrial activity (factory processes and the like) is that the mine changes daily as reserves are exploited. The shape and characteristics of the next day's reserves might differ greatly from past experience, and the current smooth-running production operation cannot necessarily be translated automatically into a smooth-running operation tomorrow.

This changed production environment is mirrored in the changed decision environment, and the mine's time-honored rules for supervisors and managers may be inapplicable with changes in the mine plan. Many decision makers may not even be aware of this fact. The systematic processes set out in the balance of this chapter are designed to forestall this problem, and some of the short-form evaluation and decision techniques are subject to additional scrutiny in Chapter 13.

Simplified Operation and Reduced Complexity

Mines that have been in operation for some time develop a very sophisticated corporate culture--a set of written and unwritten rules that evolve over time. It is these rules and procedures that allow the efficient transmission of knowledge throughout the workforce. Because it evolves over time, this tacit knowledge is commonly quite unrecognized.

The value of this knowledge becomes abruptly apparent when a team of "trained" personnel engages in a technically similar task that is not similar in economic terms. The early stages of a mine represent an archetypal situation. In this case the less-than-smooth interactions among untrained personnel (or personnel trained under an inappropriate set of rules) can impact production efficiency in dramatic ways. Production from large manufacturing plants, as well as from most capital-intensive enterprises, is quite consistent from day to day, but all mines change from one day to the next. It is very difficult to establish enough consistency for new personnel to understand

the operation well enough to operate efficiently.

New projects should be designed to allow time for these tacit rules to evolve; during this initial learning period, a guideline of "the simpler the better" should be the order of the day.

Mining Method That Reflects Corporate Philosophy
Just as day-to-day operational work is subject to tacit knowledge that is seldom consciously understood, so too is decision making at a higher level in the corporate structure.

> **Example 3.1:**
> In the late 1970s and early 1980s, many of the world's large oil companies entered the coal-mining business seeing it as an extension of their existing "energy" businesses. However, by the mid-1990s, almost all of these companies had sold out of their coal interests after failing to assimilate them into their larger corporate structures. Although the two businesses supply markets with many similarities, the production side of the mining business--where most difficult decisions are made--is quite different from that of the oil business, where production is relatively less important. The relative sizes of the businesses, financing and accounting inconsistencies, and the less fungible nature of coal compared to oil were just three other substantial differences that made this transition difficult.

In a marketplace subject to change, success that depends on decision-making consistency goes to two kinds of organizations. In the first kind, new developments should be consistent with the philosophy and capital structure of the organization. Alternatively, the organization itself must enjoy a corporate culture that can assimilate rapid change. In the absence of these conditions, new developments must be approached with caution. Mining schemes that allow sufficient time for adaptation (in the kinds of decisions being asked of senior management) should take preference over superficially more efficient methods that demand more sophisticated, higher-level decision support.

Comparing alternatives based on simplistic economics (e.g., lowest capital cost, lowest operating cost) is no longer

sufficient. In a changing world environment, the key to maintaining and improving the economics of mining lies in schemes that are adaptable enough to accommodate a wide range of circumstances and for which cost-efficiency can be sustained over this range of circumstances. The economic trade-off must incorporate average and marginal costs, as well as uncertainty and risk, and must be understandable by production and operations personnel to be a useful guideline for all of their decision making.

THE SYSTEMATIC PLANNING PROCESS

Once a proposition to use a particular mining system has been advanced, the mine design must correctly reflect the economics of this type of scheme.

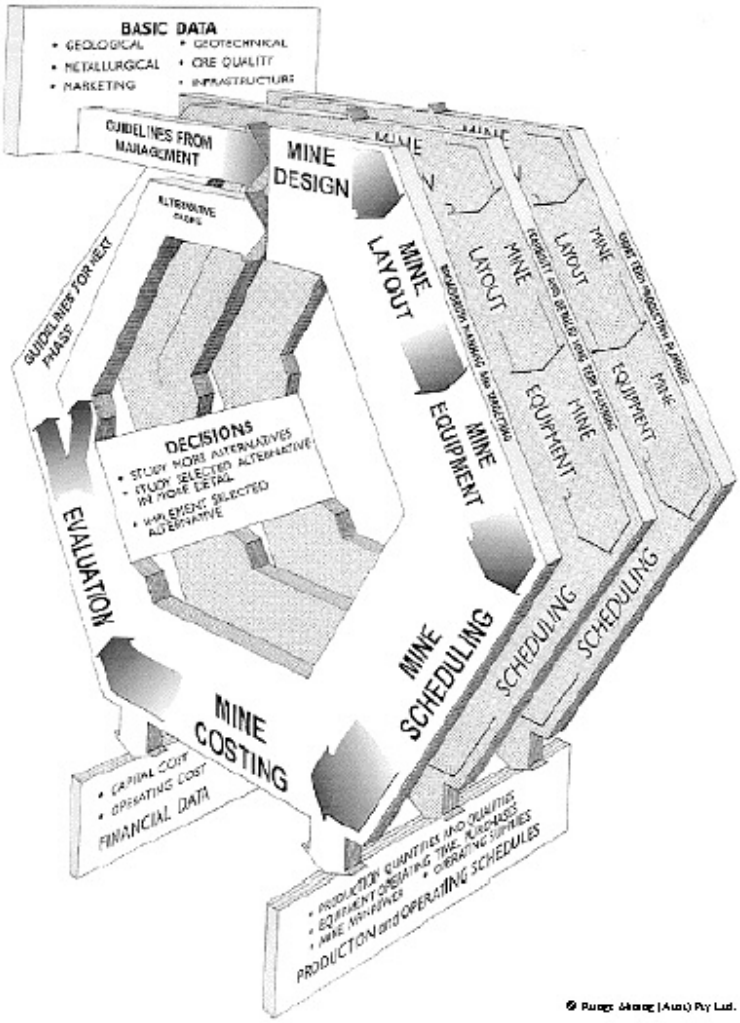
Example 3.2:

In open pit coal mining, dragline mining schemes require the mine to be planned out in long, narrow strips. The so-called technical explanation of this constraint is that a dragline can transport the waste only a limited distance. However, on examination this is not a technical constraint at all. Draglines can transport material very long distances by rehandling material--multiple numbers of times if necessary. The constraint is actually an economic one. The cost of dragline waste removal increases dramatically once material must be moved a distance exceeding twice the operating radius of the machine. A small amount of rehandling is permissible, but once a dragline scheme requires more extensive rehandling, other mining methods are usually more economic.

For loader/truck methods, different constraints apply. Unlike with the dragline, actual transport distance or the shape of the mine is seldom a limiting factor; however, the efficiency of using trucks is very sensitive to the change in elevation through which the load must be carried. The cost of truck haulage increases substantially as the open pit is deepened. This usually constrains the mine layout to match haul distances to the required change in elevation. It also places other constraints on the positioning of dump stations and out-of-pit waste dumping areas, and it governs the pit shape for optimum

placement of in-pit refill.

If there is no mine plan yet, then the costs of mining or relative costs of alternative mining schemes are unknown, so how can the mine be planned out to reflect these costs? The answer to this dilemma is that the mine-planning process is an iterative one. The expected structure of the mining costs for any one type of mine design is derived from a previous, less detailed plan of the mine or a similar mine. The first broad-brush plans of the mine and each successive iteration of the planning cycle derive and refine the rules of evaluation for subsequent phases in the cycle. Original rules derive from established corporate experience and from generalized rules of thumb such as described in this text. The systematic planning process is set out in a three-dimensional flowchart in Figure 3.1. The process consists of a similar series of steps in each phase, each undertaken in the same order. Three such phases are illustrated in Figure 3.1; however, in practice there may be any number of phases. The same series of steps are undertaken in varying amounts of detail, depending on the precision, economic action, or decision being sought.



Source: Figure has been derived from work undertaken by the author at Runge Mining (Aust) Pty Ltd., and permission to include this diagram from this source is acknowledged. Runge (1996) outlines a more comprehensive generalized model of decision making under uncertainty, of which this figure represents a narrow mining-related subset.

FIGURE 3.1 **Iterative mine-planning flowchart**

The kinds of decisions and studies through the various phases of the planning cycle are set out in Table 3.1.

TABLE 3.1 **Phases of the planning cycle**

Phase	Economic Action, Decision, or Study
First review of a new project (a broad-brush mine plan)	Use rules of thumb based on industry experience to ascertain the likely project economics. The fundamental economics of most mines are set in place by no more than four or five factors, which usually are depth, waste:ore ratio, ore characteristics (type, grade, plant yield), the selling price, and the distance to the market. Identify any particular production, market, or other constraints or risks that could potentially impact the project revenue.
Strategic planning and development strategy	This is the key phase of any project. It is the phase when critical decisions are made concerning development strategy and alternatives with substantial differences in economics are eliminated. Following this phase the fundamental economics of the mining operation are largely fixed. Changes in strategic direction have major influences on costs, risks, and capability to accommodate change. Once a strategy has been adopted, subsequent work (the phases following this step) results in more evolutionary influences on costs and risks. Evolutionary change may not result in large changes in costs of production, but it may still result in significant change in profitability. Further, the cumulative effect of extensive evolutionary change over time may permit change in broad strategy, with substantial impact on costs of production. The scope for such cumulative evolutionary change is also a part of the strategic planning phase of evaluation.
Detailed long-term mine plan or feasibility study	This phase involves at least two different types of economic evaluation: 1. Within each cost center or definable activity, the optimum mining scheme has to be determined (e.g., for waste removal, the proportions of waste moved by trucks, cast blasting, or contractor). This type of economic evaluation is undertaken by discounted average cost calculation or similar techniques restricting inputs to those factors likely to influence the result. 2. For whole-mine evaluation, the complete operating and capital costs for the mine must be developed and tabulated—leading to the whole-mine cash flow analysis.
Equipment selection	Optimization of equipment sizing and trade-offs between different machines must be undertaken on a relative cost basis—usually using some form of discounted cash flow analysis. Frequently equipment selection also involves the comparison of mining alternatives whereby some equipment purchases are delayed (either to save initial capital or because there is insufficient working room available in the early stages of mine life). These evaluations involve alternative mine schedules.
Mine development phase	In the mine development phase, most of the decisions influencing the economics are complete. This phase involves setting up the systems to compare actual costs with planned costs and to quickly highlight anomalies. The process also involves auditing of accounting procedures to ensure that precedents established for tracking of costs do not result in "day 1" decisions that are then not discovered in subsequent accounting within the mine.
Yearly planning	This is normally undertaken by using costs already established from prior experience.
Monthly planning	This is normally undertaken by using costs already established from prior experience.
Weekly and daily planning	Guidelines prepared from more exhaustive general studies should be summarized to allow field personnel to make day-to-day decisions based on economic criteria.

Figure 3.2 focuses on the evaluation task. This task applies for every phase of the process. The evaluation task has a rule-related element and an action element. The rule-related element does the following:

- Confirms that the rules of evaluation preselected for this phase are consistent with the findings from the phase. For example, if alternatives are to be primarily compared on the basis of their cost of production, the rule-related element confirms that the cost of production is a primary factor that differentiates one alternative from another.

- Establishes, for the alternative that has not been eliminated, a refined set of rules (guidelines for the next phase). This refined set of rules is the starting point for comparison among subsets of the noneliminated alternative in the subsequent phase of evaluation.

The courses of action open after any evaluation are

1. Implement the proposal.
2. Examine the proposal in more detail. Proceed to the next phase of study, seeking more reliability in the estimates prior to making a decision.
3. Examine more alternatives--either at this same level of precision or by stepping back to a more broad-brush level.
4. Abandon the project.

ECONOMIC DATA AT EACH PHASE OF THE PLANNING CYCLE

Progressing from one phase of evaluation to the next is not simply a matter of increased precision in the result--each phase also eliminates from contention alternatives that might point the mine in some entirely different direction. For this reason, the rules for evaluation during the initial broad-brush and strategic planning stages must be subject to scrutiny at the highest levels. It is at this stage when the greatest number of alternatives are available and it is easiest to incorrectly eliminate entire classes of potential plans.

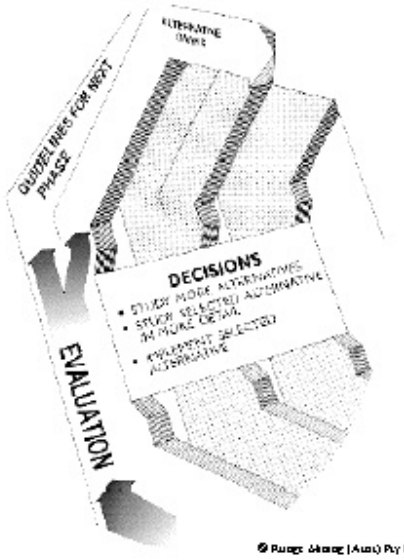


FIGURE 3.2 **Decision alternatives in the planning process**

Broad-Brush Planning Phase

The first phase of evaluation, broad-brush planning, is aimed at providing the guidelines for exploration. It is not uncommon for hundreds of potential mining scenarios to be examined at this stage--and appropriately so, since the cost of examining hundreds of alternatives at any greater level of precision may be prohibitive.

It is customary to speak of "precision" or "reliability in cost estimates" in terms of being within some percentage range of the actual value. For broad-brush analyses, the reliability of the result could easily be correct to within *&##177;* 20%, but this figure is misleading because it would be true only under the assumption that the mine plan can in fact be implemented. The validity of this assumption is demonstrated only after the detailed feasibility study.

Nevertheless, for comparing alternatives, broad-brush plans provide reliable guidelines for the likely economic characteristics of the mine--assuming such plans can be implemented. Assuming no systematic bias or omissions, a whole-mine cost estimate may be within *&##177;* 20% even if the individual inputs are accurate to only *&##177;* 40%. Individual estimates that are high can balance the estimates

that subsequently turn out to be low.

Strategic Planning Phase

The second phase of the planning cycle, the strategic planning phase, fills a critical role in optimizing the mine. It sets in place the fundamental economic structure for the entire mine. The priority in this phase of the planning cycle is to ensure three things:

- Understanding of what characterizes this (potential) mine compared to other mines or what characterizes this mine's products as judged by the customer--and adoption of a strategy that maximizes this value. Even if the fundamental cost structure is not different from that of other mines, there may be significant points of differentiation from other mines. Differentiation provides the basis for premium pricing of products, sales of products when other mines cannot attract sales, or scope for adaptation as the mine is developed or as customer needs change. Most of these factors are intangible and do not show up on cash flow statements.
- Development of the mine itself with sufficient scope for change. Most mines are developed around a selected base case, and the equipment sizing and development strategy are optimized for the base case estimate of market demand. Historically, this base case output usually proves to be too low in the long term (most mines expand from their initial production rate) but also too optimistic in the early stages of the mine life. With traditional investment decision methods, it is difficult to avoid adopting some base case production no matter how uncertain one may be of market interest. Planning and decision making have to be based on something! Nevertheless, the inevitability of change must be acknowledged. The strategy must plan for, and management must immediately start the implementation of processes leading to, changes in the mine in line with changes in the external world.
- The adopted strategy that produces the preceding two results will yield a plan with a certain set of economic characteristics (cost structure, risk, etc.). The third objective of the strategic planning process is

to understand which of these characteristics impacts the mine economics the most--as a guideline for following stages of the planning. Costs may then be optimized for those elements of the mine plan that are the most controllable.

Figure 3.3 shows in probabilistic terms the objective of mine planning during this strategic planning phase. Strategic planning is primarily concerned with changing, rejecting, and tentatively adopting choices that are fundamentally different in expected value terms. The focus is on choices that have the highest expected return, rather than on choices that are less risky. Although Figure 3.3 presents these distributions as if they were normal distributions, in practice the dispersion of potential outcomes, even if it were known, is unlikely to fit any standard statistical shape. This does not invalidate the process--it merely recognizes the subjective nature of decision making with gross shortcomings in information. Thus, "expected value" in this context means the most likely outcome, or (in the case of a normal distribution) the mean of the distribution. Alternatives compared by using this focus of valuation are tentatively accepted on the following assumptions:

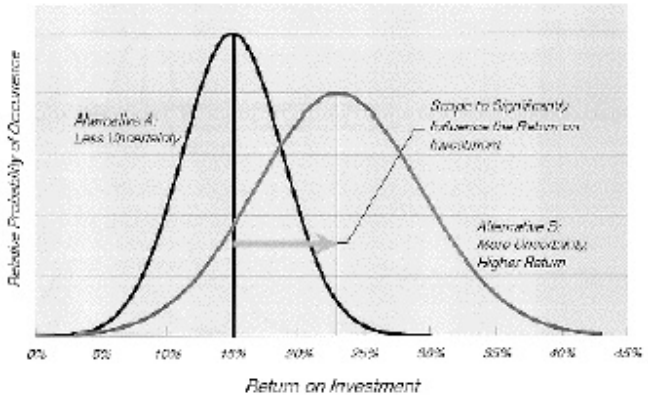


FIGURE 3.3 **Return on investment during the strategic planning phase**

- Uncertainties (the spread of potential outcomes) can be reduced in subsequent evaluation phases.
- Subsequent evaluation will not significantly change the expected value of the distribution as uncertainties are resolved.

Once the strategic planning phase is complete, the overall expected return on investment seldom changes in any significant way. Subsequent phases of the planning are

concerned with refining the plan.

If a detailed feasibility study has been completed and the project does not measure up to the required return on investment, then further detailed study is not the course of action to follow. A significant change in the return on investment requires some substantial or fundamental change in the strategy for development, not just refinement of some existing strategy.

> **Example 3.3:**
> At the start of many open pit gold projects, the mine is very shallow, grades are low, and mining costs as a proportion of overall project costs are very small. Most cost is in the ore treatment because of the large volumes of ore that must be handled. The priority for planning rests on ensuring delivery of the expected grade to the mill, even at the expense of mining costs. Any excess dilution will decrease throughput of ore proportionally and reduce revenue. The primary forms of technology a gold company needs at this stage are skills in ore treatment.
>
> If, however, this first plan does not yield sufficient return, rather than fine-tuning the treatment process, the alternative may be to adopt an entirely different strategy. Development as an underground mine may allow much higher grades earlier in the mine life. In this alternative scenario, treatment costs may be the minor part of the mine cost structure. Mining costs will be more significant, and grade control will assume a lower priority than, say, reliability of ore delivery. Scheduling constraints, a common issue in underground mining, may be another priority. The primary forms of technology a gold company needs for this scenario are skills in underground mine development.

If a reexamination of the strategy is needed, it may be difficult to achieve with an in-house team whose skills have already been selected with an existing strategy in mind. The skills needed to understand alternative strategies (and implement them) may be completely different.

The strategic issues in a mine development can be broadly grouped in three ways, as shown in Table 3.2.

TABLE 3.2 **Strategic issues in mine development**

Strategic Factor	Example
Key strategic factor is affecting revenue	▪ Selling price of product, including premiums or discounts for quality characteristics ▪ Quantity and quality of product produced ▪ Quantity of product able to be sold, including transport constraints and factors likely to disrupt mine output ▪ Start-up scheduling problems in the mine or mill delaying initial deliveries
Key strategic factor is affecting cost	▪ Surface mines: Waste removal costs, rehabilitation costs ▪ Underground coal mines: Development to establish primary workings, roof support, ventilation costs ▪ Underground hard-rock mines: Development and access costs, drilling and blasting costs, backfilling costs, ground support ▪ Quarries: Fragmentation, environmental (blasting) constraints
Key strategic factor is affecting risk	▪ Uncertainty in orebody definition (more critical in metalliferous mining, underground coal mines, and deeper mines) ▪ Price and currency volatility, as well as constraints associated with the financial structure limiting the scope for restructuring in the face of change ▪ Technology, as well as the potential for competitors to adopt new technology while the company's mine is constrained to use older, less economic technology ▪ Industrial disruption, as well as the constraints on change caused by institutionalized industrial agreements (when competitors are not so constrained) ▪ Environmental uncertainties (frequently the risk is not the size of these impacts, but rather what additional costs are incurred because of delays) ▪ Political uncertainties: Access to land, community support

One of the key requirements of the strategic planning phase is to understand the risk characteristics of the particular mining method chosen. Subsequent planning can work out the most appropriate way of minimizing the impact of the uncertainty on the project cash flow.

Consider, for example, decision making for design of a processing facility. The economics of such a facility are sensitive to the precision of the estimates of plant yield, and often this cannot be determined accurately at the planning stage. Three strategically different alternatives present themselves:

1. Design the plant capacity for the "expected" plant yield. This alternative means that there is a 50% chance that the plant will be overdesigned. Capital will be spent that need not be spent. There is also a 50% chance that the plant will be underdesigned. In this event output will be reduced and revenue will suffer while all other costs remain unchanged. Is there scope for change in this plant design? Loss of revenue, since it is far more important than slightly higher cost, should be avoided even if higher costs are incurred. Overtime is an option available to some mine operators; it allows maintenance of production but at

higher cost.

2. Design the plant capacity with production capacity to cover, say, 90% of the likely yield variation. This alternative uses capital to reduce risk. Compared to alternative 1, it is a decision to spend extra capital (a certain cost now) when on average there is only a 40% chance the extra capital will be needed. There is a 10% chance that the capital will still be inadequate. From an engineering viewpoint, alternatives that have a 90% chance of success are preferable to alternatives that have only a 50% or less chance of success. From an economic viewpoint, planning to address the problem by using remedial measures may be cheaper.
3. Deliberately underdesign the plant (say, at a design capacity for which there is just a 30% probability that it will be adequate). At the same time, put in place contingency plans for immediate remedial action in the likely event (70% probability) that the plant indeed proves inadequate. The philosophy in this case is that the plant is expected to be inadequate, but it is perhaps not known in which particular area the inadequacy will show up, and the costs of covering all of the potential areas are excessive. Of course, there is always a 30% chance that it won't be inadequate at all.

Are there any guidelines as to which strategic option is preferable? The primary difference relates to institutionalized response to change.

An organization that has difficulty or high costs in accommodating change must favor alternative 2. The initial premium reduces the likelihood of subsequent change. On the balance of probabilities, this alternative results in overdesign--arguably still an "error" in economic terms--but frequently correctable by further expansion (including more capital if necessary).

An organization that can readily accommodate change has the potential to avoid expenditures that will automatically be made if alternative 2 is selected. Nevertheless, remedial measures are always very evident costs and are usually regarded as signs of failure, whereas the likely overcapitalization in alternative 2

is a less evident cost.

To summarize:

1. The primary economics of the mine are largely set in place by those factors that impact the revenue. The understanding of these factors (i.e., orebody characteristics) may be under the control of the mine planner, but the factors themselves may not be. The mine must be rich enough to be viable even in the event of a low-probability outcome. Alternatively, the plan must be adaptable enough to accommodate the changes necessary to be viable despite the uncertainty of the outcome.
2. Once the factors influencing revenues are addressed, the focus of the mine design must be on optimizing those factors that impact the cost structure of the mine the most. This is the area most under the control of the mine planner.
3. Mine planning must consider the risk factors--particularly those likely to cause loss of revenue. Mines can reduce their sensitivity to many risks through higher operating costs.

Many uncertainties cannot be removed by operating and capital cost changes to the plan or even by contingency planning. In these cases the decision to proceed is based on an "expected" return on investment high enough to cover the risk.

For those factors that can be quantified in capital or operating cost terms, a decision not to adopt the higher-cost/lower-risk alternative is a decision to accept a higher risk. In project assessments, the discount rate must reflect the risk. If it does not do so, then it implicitly biases the mine plan toward higher-risk options that yield a higher return.

An important outcome of this planning phase is to understand the cost structure of the mine as a guideline (the "rules") for future mine planning. The future mine plan must include an appropriate balance of

- overall focus on the more profitable reserves
- initial stages of the mine plan focused on reserves yielding strong cash flows (lower development costs,

higher grades, etc.)

- mining schemes that facilitate change (expand, produce alternative products, etc.)

In open pit coal mining, the cost structure and related decisions are frequently determined through a cost-ranking analysis, described in more detail in Chapter 6.

In underground metal mining, a similar procedure usually involves cutoff-grade analysis and the influence of selling prices of polymetallic ores on development strategies. This is also covered in Chapter 6.

In open pit hard-rock mines, this phase of the planning process is called pit optimization. It defines the boundaries of a pit given certain conditions and the likely change in boundaries given changes in any of the inputs. This is also covered in Chapter 6.

The result of this strategic planning phase is to establish where mining is to take place, at what rate, and broadly how it is to be mined, i.e., what sequence and method. It establishes the ground rules for subsequent analysis.

Example 3.4:

Figure 3.4 shows two proportional pie charts of the cost structure of a large-scale open pit coal mine. The start of the mine, at a depth of 30 m, has relatively low costs--and 34% of these costs are in actual mining of the coal. At the end of the mine, at a depth of 100 m, the direct coal-mining costs represent only 9% of the much larger total. The dragline costs (representing 36% of the total costs) are also less significant than the costs associated with truck/shovel waste removal. At depth, for overall mine design efficiency, truck/shovel waste removal would take priority, even at the expense of dragline efficiency.

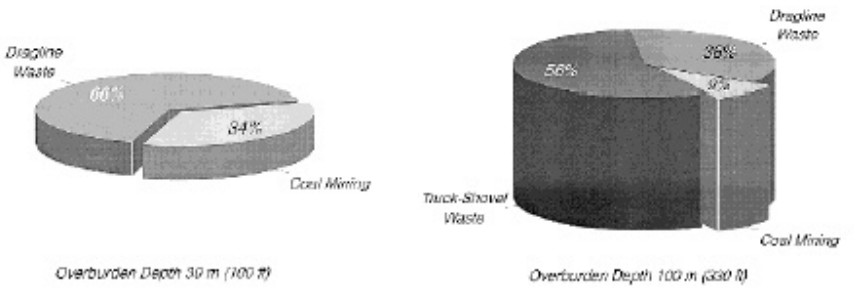


FIGURE 3.4 **Cost structure through mine life**

Subsequent Planning Phases

The contribution of the detailed and subsequent planning phases is to improve the confidence in the estimates--through understanding the implementability, constraints, and capital and operating costs. Figure 3.5 shows a probabilistic representation of the changes in return on investment during this detailed phase.

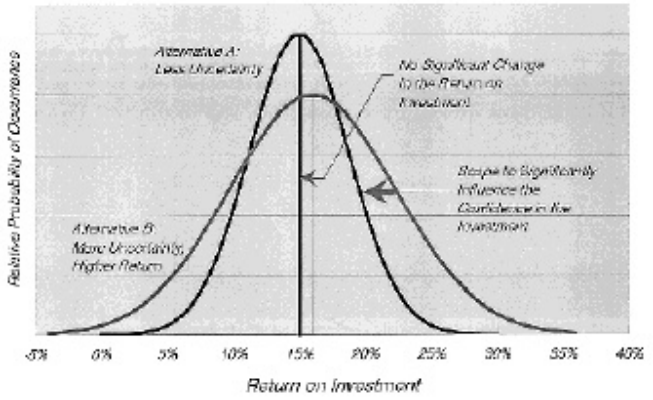


FIGURE 3.5 **Effect on precision and return on investment during the detailed phase**

Once the strategies are in place and the economics and risks associated with those strategies are clearly understood, then the details can be addressed in a long-term planning and feasibility phase. The key word for this and subsequent phases is implementability. The planning is directed at ensuring the highest probability that the production that is supposed to be achieved is in fact achieved. It also involves developing procedures so that this translation from "planned" to "actual" endures. The production techniques, costs, and cash flow characteristics will be greatly refined over those of the broad-brush phases, but it is unlikely that whole-mine economics (the

whole-mine return on investment) will be notably different than the estimates in previous phases of study.

Issues to be addressed include

- equipment selection and optimization
- mine development, particularly the detailed layout of the mine and simulation of the revenue stream for the first few years, when the cash flows are most sensitive
- optimization of bench heights, widths, access constraints, and scheduling alternatives
- yearly, monthly, and shorter-term planning
- economic guidelines for short-term operational decision making

These issues are addressed in the balance of this book.